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# ASSIGNMENT - II

1] What are characteristics of C++ Programing Language

→ They are data members of a class which contains the values

Characteristics ← Static
Characteristics ← Nonstatic

Eg. int i;
int a;

2] What is meant by a class ?

→ 1) To achieve object orientation a new data type is provided ie → 'CLASS'
2) Class is a user-defined datatype almost same as 'structure' in C programming language

Class ← Characteristics (Data Members)
Class ← Behaviours. (Functions of Class)

Eg.
```
Class Demo        ] class name
{ public:
    int a;        ]
    int b;        ] Characteristics
                  ]
    void run() {} ] Behaviour.
};                ] class
```

3] Write notes on Constructor & Destructor.

→ A] Constructor

Used to initialize characteristics & allocate resources

Types : 1] Default 2] Parameterized
3] Copy

B] Destructor

Used to deallocate resources which were allocated inside the constructor.

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Rules :

1] Name of constructor → same as class name
2] All constructor & destructors under public access specifier
3] Should be no return type for Cons & Dest.
4] If we create any object w/o passing any parameter then the default constructor gets called.
5] object created by passing parameters → Parameterized constructor is called
6] object created by passing another object, parameterized constructor get called.
7] No need of explicit call to Cons & Dest.
8] & operator used inside copy constructor.
↳ Reference operator, if not written then it may lead to recursive calls.

1] what are data type in c++ & list them with size.

| Primitive | Derived | User defined |
|---|---|---|
| char (1b) | Array [Add^n of size of data member] | Structure (Add^n of size) |
| int (4b) | pointer (8b) | Union [biggest size of member] |
| float (4b) | Functions | |
| double (8b) | Reference | Class (Add^n of chars + size of Beh) |
| void | | |
| Boolean (1 bit) | | |

2] Explain tool chain of C++ Program.
→ Tool Chain of C++ is same as Tool chain of C.

gcc chain

Programmer → Editor (Demo.cpp)
Editor → Preprocessor (Demo.i)
Preprocessor → Compiler (Demo.asm/s)
Compiler → Assembler (demo.obj)
Assembler → Linker (demo.exe)
Linker → Loader → M/P

1- Tools
2- editor & loader by OS

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6] What is meant by "Abstraction"
→ - Hiding something from outside world
To Achieve this we use access specifiers.
- ACCESS SPECIFIERS → Who can & cannot access members (char + beh) of a class

3 Access specifiers :
1] Public (All can access) ] * access.cpp
2] Private (only inside same class) ]
3] Protected (only child can & also in same class) ] * access1.cpp

- Irrespective of Access specifier type, memory for all chars & beh gets allocated.

7] What is meant by Access specifier ? explain with example.
→ Refer Q.6]
Example :
1] Public, Private & Protected

```
Class Demo
{   public:
      int i;

    private
      int a;

    Protected
      int j;

public:
Demo ()
{ i = 10;
  a = 20;
  j = 30; }
};
```

8) Write a program to find out maximum of 2 nos using procedural (C) & object oriented approach (C++)

→ C Program

```
#include <stdio.h>
int main()
{ int no1, no2, max;
  printf("Enter 2 numbers: ");
  scanf("%d %d", &no1, &no2);

  max = (no1 > no2) ? no1 : no2;
  printf("The maximum of %d & %d is: %d\n",
          no1, no2, max);
  return 0;
}
```

C++ Program

```
#include <iostream>
using namespace std;
int main()
{ int no1, no2, max;
  cout << "Enter 2 numbers: ";
  cin >> no1 >> no2;

  max = (no1 > no2) ? no1 : no2
  cout << "The max of " << no1 << "and"
  << no2 "is" : " << max << endl;

  return 0;
}
```

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Q] Explain object oriented Programming paradigm [OOP Concepts]

OOP concepts in C++ & Java

Encapsulation Abstraction Inheritance Polymorphism

1] Encapsulation :
- Binding characteristics & behaviours together
- To achieve this we create a class.

2] Abstraction :
- Hiding something from outside world
- Achieve this we use access specifier (like pvt)

3] Inheritance :
- Considered as 'reuse ability'
- Achieve this we need to inherit the derived class from parent class

Parent (base)
↓
Child (derived)

4] Polymorphism : (अनेक रूप)
- Single name multiple behaviours
a) Compile time - overloading
b) Run time - overriding

10) Write down the difference between C and C++

| Aspect | C | C++ |
|---|---|---|
| Paradigm | Procedural PL | Object oriented PL |
| Primary Use | System Level Programming (Eg OS, drivers) | System & Application level programming (e.g. games, GUIs) |
| Class | Not supported | Fully supports classes & objects |
| Functions | must be declared before use | can be defined anywhere. |
| Data Security | Limited (no direct mechanism for data hiding) | Enhanced with encapsulation & Access specifier. |
| Memory management | Uses : malloc(), calloc() & realloc(), | Uses new & delete operators for dynamic memory management |
| Except Handling | Not Supported | Supported using - try, catch and throw. |
| File extension | .c | .cpp |